

Dashboard Overview & System Architecture

The **Medical Insurance Applied Statistical Modeling & Interactive Dashboard** bridges formal mathematical inference with an accessible, user-driven software tool. Developed in Python using **Streamlit** and **Statsmodels**, the application translates raw demographic and biometric records into real-time statistical insights, hypothesis tests, and predictive regression analytics.

The dashboard is structured into three distinct functional modules:

Tab 1: Exploratory Data Analysis (EDA)

- **Interactive Filtering:** A persistent sidebar allows dynamic filtering across patient age ranges, smoker status, and geographic regions.
- **Descriptive Summary:** Computes key parametric and non-parametric summary metrics—mean, median, standard deviation, interquartile range (IQR), skewness, and kurtosis—reacting instantly to filtered subsets.
- **Interactive Visualizations:** Renders distribution histograms and boxplots of medical charges, scatter plots exploring BMI interactions, and a bivariate correlation heatmap using Plotly.

Tab 2: Hypothesis Testing Lab

- **Two-Group Comparison Engine:** Evaluates whether medical expenses differ significantly between smokers and non-smokers.
- **Automated Normality & Variance Verification:** Performs Shapiro-Wilk tests for distribution normality and Levene's test for equality of variance.
- **Adaptive Test Execution:** Automatically routes to a standard Two-Sample t -test if assumptions are met, or executes the non-parametric Mann-Whitney U test if normality is violated.
- **Categorical & Multi-Group Analysis:** Features interactive dropdowns allowing users to inspect either:
 - A **Chi-Square Test of Independence** assessing geographic links to smoking.
 - A **One-Way ANOVA (F -test)** assessing charge variance across regional categories.
- **Dynamic Inference:** Displays computed test statistics, exact p -values, and color-coded decision verdicts (Reject vs. Fail to Reject H_0 at $\alpha = 0.05$).

Tab 3: Live Prediction & Diagnostic Engine

- **OLS Multiple Linear Regression:** Fits an Ordinary Least Squares model estimating medical charges based on age, sex, BMI, children, smoking status, and region.
- **Real-Time Cost Estimator:** Enables users to manipulate individual patient attributes (sliders and selectboxes) to calculate expected charges with a 95% Confidence Interval in real time.
- **Gauss-Markov Diagnostics:** Validates core regression assumptions by generating:
 - A **Residuals vs. Fitted Values** scatter plot to verify linearity and homoscedasticity.
 - A **Normal Q-Q Plot** alongside the Jarque-Bera metric to inspect error-term normality.

- **Variance Inflation Factor (VIF)** tables across continuous predictors to screen for severe multicollinearity.

1. Descriptive Metrics

Sample Mean ($\bar{x}$)

- **Definition:** The arithmetic average of all observed values in a dataset, representing its central balance point.
- **Formula:**

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

- **Variables:**
 - x_i : Value of the i -th individual observation.
 - n : Total sample size (number of observations).
 - $\bar{x}$: Calculated sample mean.

Sample Standard Deviation (s)

- **Definition:** The measure of average dispersion or Euclidean scatter of individual data points around the sample mean.
- **Formula:**

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

- **Variables:**
 - x_i : Value of the i -th observation.
 - $\bar{x}$: Sample mean.
 - $n - 1$: Degrees of freedom (Bessel's correction for sample variance).
 - s : Sample standard deviation.

Interquartile Range (IQR)

- **Definition:** The statistical spread spanning the central 50% of an ordered distribution, unaffected by extreme outliers.
- **Formula:**

$$\text{IQR} = Q_3 - Q_1$$

- **Variables:**

- Q_3 : Third quartile (75th percentile of the ordered dataset).
- Q_1 : First quartile (25th percentile of the ordered dataset).

Sample Skewness (g_1)

- **Definition:** The third standardized central moment that quantifies the directional asymmetry and tail-weight of a distribution.
- **Formula:**

$$g_1 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{\left[\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right]^{3/2}}$$

- **Variables:**
 - x_i : Value of the i -th observation.
 - $\bar{x}$: Sample mean.
 - n : Total sample size.
 - g_1 : Skewness coefficient (> 0 indicates right-skewed/heavy positive tail).

Sample Kurtosis (a_4)

- **Definition:** The fourth standardized central moment evaluating the "tailedness" and propensity for extreme outliers relative to a Gaussian distribution.
- **Formula:**

$$a_4 = \frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^4}{\left[\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \right]^2}$$

- **Variables:**
 - x_i : Value of the i -th observation.
 - $\bar{x}$: Sample mean.
 - n : Total sample size.
 - a_4 : Kurtosis value (= 3 for a normal distribution; > 3 indicates heavier tails / leptokurtic).

2. Inferential Hypothesis Testing

Shapiro-Wilk Normality Test (W)

- **Definition:** Evaluates how closely the ordered sample observations align with the expected order statistics of a standard normal distribution.
- **Formula:**

$$W = \frac{(\sum_{i=1}^n a_i x_{(i)})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

- **Variables:**
 - $x_{(i)}$: The i -th ordered sample value (order statistic: $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$).
 - $\bar{x}$: Sample mean.
 - a_i : Weights computed from the covariance matrix and expected values of normal order statistics.
 - W : Test statistic bounded between 0 and 1 (values close to 1 indicate normality).

Levene's Test for Equality of Variances (W)

- **Definition:** Tests the equality of variances across k groups based on absolute deviations from group medians.
- **Formula:**

$$W = \frac{N - k}{k - 1} \cdot \frac{\sum_{i=1}^k N_i (\bar{Z}_{i.} - \bar{Z}_{..})^2}{\sum_{i=1}^k \sum_{j=1}^{N_i} (Z_{ij} - \bar{Z}_{i.})^2}$$

- **Variables:**
 - k : Total number of comparison groups ($k = 2$ for smokers vs. non-smokers).
 - N : Total number of observations across all groups.
 - N_i : Sample size of group i .
 - $Z_{ij} = |x_{ij} - \tilde{x}_i|$: Absolute deviation of observation j in group i from group median $\tilde{x}_i$.
 - $\bar{Z}_{i.}$: Mean of Z_{ij} within group i .
 - $\bar{Z}_{..}$: Grand mean of all Z_{ij} values.

Independent Two-Sample Welch's t -Test (t)

- **Definition:** Parametric test measuring whether the true population means of two independent groups differ significantly without assuming equal variances.
- **Formula:**

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

- **Variables:**
 - $\bar{x}_1, \bar{x}_2$: Sample means of group 1 and group 2.
 - s_1^2, s_2^2 : Sample variances of group 1 and group 2.
 - n_1, n_2 : Sample sizes of group 1 and group 2.

- t : Calculated t -statistic compared against a Student's t -distribution.

Mann-Whitney U Test (U)

- **Definition:** Non-parametric rank-sum test comparing whether random observations from one group tend to exceed observations from another group.

- **Formula:**

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - R_1, \quad U = \min(U_1, U_2)$$

- **Variables:**

- n_1, n_2 : Sample sizes of group 1 and group 2.
- R_1 : Sum of ranks assigned to observations belonging to group 1 after pooling and ranking all combined observations.
- U : Final test statistic used to evaluate the p -value.

Chi-Square Test of Independence (χ^2)

- **Definition:** Tests whether two categorical variables are statistically independent by measuring discrepancies between observed and expected frequencies.

- **Formula:**

$$\chi^2 = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \quad \text{where} \quad E_{ij} = \frac{R_i \cdot C_j}{N}$$

- **Variables:**

- O_{ij} : Observed count in row i , column j of the contingency table.
- E_{ij} : Expected theoretical count under the null hypothesis of independence.
- R_i : Total observations in row i (marginal row total).
- C_j : Total observations in column j (marginal column total).
- N : Total sample size.
- r, c : Number of rows and columns (Degrees of freedom $df = (r - 1)(c - 1)$).

One-Way ANOVA (F -Statistic)

- **Definition:** Tests whether population means across three or more groups differ by calculating the ratio of variance between group means to variance within groups.

- **Formula:**

$$F = \frac{MSB}{MSW} = \frac{\frac{SSB}{k-1}}{\frac{SSW}{N-k}}$$

- **Variables:**

- $SSB = \sum_{j=1}^k n_j (\bar{x}_j - \bar{x})^2$: Between-group sum of squares.
- $SSW = \sum_{j=1}^k \sum_{i=1}^{n_j} (x_{ij} - \bar{x}_j)^2$: Within-group sum of squares.
- k : Total number of categorical groups.
- N : Total number of observations across all groups.
- MSB,MSW: Mean square between and mean square within.

3. Multiple Linear Regression & Parameter Estimation

Ordinary Least Squares (OLS) Parameter Vector ($\hat{\beta}$)

- **Definition:** The vector of linear weights that minimizes the total sum of squared prediction errors.
- **Formula:**

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}$$

- **Variables:**
 - $\mathbf{X}$: The $n \times (k + 1)$ design matrix of explanatory variables (including a leading column of 1s for the intercept).
 - $\mathbf{Y}$: The $n \times 1$ vector of observed dependent values (charges).
 - $\hat{\beta}$: The $(k + 1) \times 1$ vector of estimated regression coefficients $(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k)$.

Coefficient of Determination (R^2)

- **Definition:** The proportion of total variance in the dependent variable explained by the regression model.
- **Formula:**

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

- **Variables:**
 - y_i : Actual observed value of the i -th target observation.
 - $\hat{y}_i$: Model-predicted (fitted) value for observation i .
 - $\bar{y}$: Mean of the observed target values.
 - SS_{res} : Residual sum of squares (unexplained variation).
 - SS_{tot} : Total sum of squares (total variation around the mean).

Adjusted R^2 ($\bar{R}^2$)

- **Definition:** A penalized version of R^2 that adjusts for the number of explanatory variables relative to sample size.
- **Formula:**

$$\bar{R}^2 = 1 - \left[\frac{(1 - R^2)(n - 1)}{n - k - 1} \right]$$

- **Variables:**
 - R^2 : Standard coefficient of determination.
 - n : Total number of observations.
 - k : Number of independent predictor variables (excluding the intercept).

4. Gauss-Markov Residual Diagnostics

Jarque-Bera Test for Residual Normality (JB)

- **Definition:** An asymptotic goodness-of-fit test assessing whether model residuals match Gaussian skewness (0) and kurtosis (3).
- **Formula:**

$$JB = \frac{n}{6} \left(S^2 + \frac{(K - 3)^2}{4} \right)$$

- **Variables:**
 - n : Number of residuals (sample size).
 - S : Skewness of the residuals ($e_i = y_i - \hat{y}_i$).
 - K : Kurtosis of the residuals.
 - JB : Test statistic distributed as χ^2 with 2 degrees of freedom.

Variance Inflation Factor (VIF_j)

- **Definition:** Quantifies how much the variance of an estimated regression coefficient is inflated by linear dependence with other explanatory features.
- **Formula:**

$$VIF_j = \frac{1}{1 - R_j^2}$$

- **Variables:**
 - R_j^2 : Coefficient of determination obtained by regressing predictor X_j onto all remaining independent predictors.
 - VIF_j : Variance inflation factor for the j -th predictor ($VIF_j > 5$ indicates high collinearity).

Population Cohort Controls

Age Horizon

1864

BMI Horizon

15.9653.13

Smoking Status

yesno

Regions

southwestsoutheastnorthwestnortheast

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Econometric & Biostatistical Inference Engine

Applied Statistical Modeling, Gauss-Markov Validation, and Parametric Diagnostics

Empirical ExplorationInferential Hypothesis TestingOLS Modeling, Transformations & Diagnostics

Sample Cohort1,338 records

Mean Cost\$13,270.42

Cost IQR (Dispersion)\$11,899.63

Fisher Skewness1.52

Empirical Moments Summary

Metric	Mean	Std Dev	Median	IQR	Skew	Kurtosis
age	39.21	14.05	39.00	24.00	0.06	-1.25
bmi	30.66	6.10	30.40	8.40	0.28	-0.05
children	1.09	1.21	1.00	2.00	0.94	0.20
charges	13270.42	12110.01	9382.03	11899.63	1.52	1.61

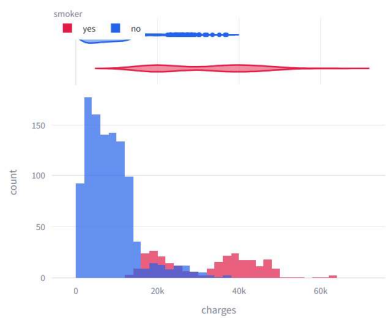
Density Distribution & Right-Tail Infiltration



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Density Distribution & Right-Tail Infiltration



Bivariate Correlation Heatmap



Automated Hypothesis Verification (alpha = 0.05)

Test 1: Two-Group Comparison with Assumption Verification

Stratification Factor: smoker

Continuous Target Metric: charges

Applied Engine: Mann-Whitney U Rank-Sum Test (Non-Parametric Fallback)

Calculated Test Statistic: 284,133.0000

p-value: 5.2702e-130

Significant

Assumption Diagnostics:

- Normality (yes): $p = 3.62e-99$
- Normality (no): $p = 1.45e-28$
- Levene's Equal Variance: $p = 1.56e-66$

Statistical Decision: Reject H0. Significant distributional divergence across smoker tiers.

Test 2: Categorical Independence or Multi-Factor Variance

Execute:

Chi-Square Test of Independence One-Way ANOVA with Post-Hoc Tukey HSD

Factor A: smoker, northwest, northweat, southeast, southwest

Manage app

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Empirical ExplorationInferential Hypothesis TestingOLS Modeling, Transformations & Diagnostics

Statistical Specification & Diagnostics

Dependent Target Formulation (Y):

Raw Charges (\$)Log-Transformed charges: log(charges)

Model Structure:

Additive ModelInteraction Model (BMI × Smoker)

Adjusted R²

AIC

BIC

F-Test p-value

0.7494

27,113.5

27,160.3

0.00e+00

Gauss-Markov Assumption Health Auditor

Homoscedasticity (Breusch-Pagan):

Residual Normality (Jarque-Bera):

Multicollinearity (Max Continuous VIF):

VIOLATION

VIOLATION

PASS

BP p-value: 1.45e-22

JB p-value: 7.86e-157

Max VIF: 1.81 (Safe threshold < 5.0)

Residuals vs. Fitted

Normal Q-Q Plot

Cook's Distance (Influence Analysis)

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Gauss-Markov Assumption Health Auditor

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Residuals vs. Fitted

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> View Full OLS Regression Table

Real-Time Inference & Interval Estimation

Patient Age

32

Body Mass Index

27.50

Smoker Tier

no

Point Estimate

\$4,992.82

Sex Tier

male

Dependents Count

1

Region Tier

southwest

95% CI (Mean Response): [4,288.41, 5,785.23]

95% PI (Individual Response): [-6,825.88, 16,911.52]

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